

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE CODE: CSA51 **COURSE NAME:** Cryptography and Network Security

COURSE OUTCOME COVERED

CO4: *Implement best security practices for IP, email, and web service using PGP, S/MIME for enhancing email Security.CO (BL3, BL6)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

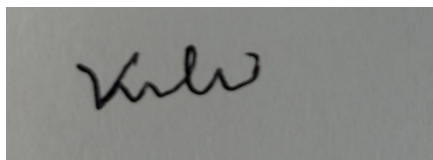
TOTAL MARKS: 25

Annexure A Industry Problem- Solving Task

Code of Conduct:

I _____VARSHA.S/192511358_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A rectangular box containing a handwritten signature in black ink. The signature appears to be 'Varsha.S' written in a cursive, stylized font.

Annexure A

1. SSL/TLS Overhead and Web Server Performance Optimization

A web server uses TLS 1.3 for secure communication. Each HTTP response carries 2000 bytes of payload. TLS adds 25 bytes (record header + authentication tag) and handshake overhead of 3 KB per session. The server handles 10,000 requests per second, with 1,000 new TLS handshakes per second.

Task:

- A. Calculate the total data transmitted per second including TLS overhead.
- B. Determine the percentage overhead due to TLS encryption.
- C. Estimate bandwidth consumed due to handshake overhead per second.
- D. Analyze the trade-off between security and latency, and propose optimization techniques (e.g., session resumption, HTTP/2, or QUIC).

2. PGP-Based Email Encryption Load in Enterprise System

An enterprise email server encrypts emails using PGP. Each email contains a 10 MB attachment. PGP introduces 800 bytes overhead for key encryption and signatures. The system processes 5,000 emails per day.

Task:

- A. Calculate the total data transmitted per day including overhead.
- B. Determine the percentage increase in storage due to PGP.
- C. Analyze the computational overhead during encryption/decryption.
- D. Recommend optimization strategies (e.g., compression, batching, or hardware acceleration).

3. IPSec Transport vs Tunnel Mode Performance Comparison

A company compares IPSec transport mode and tunnel mode for securing internal communications. Original packet size = 1000 bytes payload + 20-byte header. Transport mode adds 30 bytes, while tunnel mode adds 50 bytes overhead. Traffic volume = 20,000 packets per second.

Task:

- A. Compute total packet size for both modes.
- B. Calculate bandwidth usage for each mode per second.
- C. Compare efficiency and security benefits of both modes.
- D. Recommend the appropriate mode for internal vs external communication with justification.

4. S/MIME Certificate Management in Large Organization

An organization manages 12,000 employees using S/MIME certificates. Each certificate is 2 KB, and certificate renewal happens every year. During renewal, both old and new certificates are stored simultaneously for 1 month.

Task:

- A. Calculate total storage required during the renewal period.
- B. Estimate certificate distribution overhead across the network.
- C. Analyze challenges in certificate lifecycle management at scale.
- D. Propose optimization strategies (e.g., automation, centralized PKI, or short-lived certificates).

5. Email Encryption Latency in Cloud-Based Systems

A cloud email service encrypts outgoing emails using AES (data) and RSA (key exchange). AES encryption takes 0.2 ms per MB, and RSA key exchange takes 5 ms per email. Each email has 8 MB attachment. The system processes 8,000 emails per hour.

Task:

- A. Calculate total encryption time per email.
- B. Estimate total processing time per hour.
- C. Analyze the performance bottleneck between symmetric and asymmetric operations.
- D. Recommend optimization strategies (e.g., ECC adoption, parallel processing, or hardware acceleration).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

1. SSL/TLS Overhead and Web Server Performance Optimization

Given:

- HTTP Payload = **2000 bytes**
- TLS overhead per record = **25 bytes**
- TLS handshake overhead = **3 KB = 3072 bytes**
- Requests = **10,000 requests/sec**
- New TLS handshakes = **1,000 handshakes/sec**

A. Total Data Transmitted per Second

Data per HTTP Response

= Payload + TLS overhead

= 2000 + 25

= **2025 bytes**

For **10,000 requests/sec**

= $2025 \times 10,000$

= **20,250,000 bytes/sec**

Handshake overhead

= 3072×1000

= **3,072,000 bytes/sec**

Total Data

= 20,250,000 + 3,072,000

= **23,322,000 bytes/sec**

≈ **23.32 MB/s**

B. Percentage TLS Overhead

Total TLS overhead

= Record overhead + Handshake overhead

= $(25 \times 10,000) + (3072 \times 1000)$

= 250,000 + 3,072,000

= **3,322,000 bytes**

Percentage overhead

= **14.24%**

C. Handshake Bandwidth per Second

Handshake overhead

= **3,072,000 bytes/sec**

≈ **3.07 MB/s**

D. Analysis and Optimization

Trade-off

Advantages

- Strong encryption
- Data confidentiality
- Authentication
- Integrity protection

Disadvantages

- Extra bandwidth
- Handshake latency
- Increased CPU utilization

Optimization Techniques

- **TLS Session Resumption** reduces repeated handshakes.
- **HTTP/2** multiplexes requests over one TLS connection.
- **QUIC/HTTP3** eliminates TCP handshake delays.
- Enable **Keep-Alive** connections.
- Use **TLS 1.3 Zero Round Trip (0-RTT)** for returning users.

2. PGP-Based Email Encryption Load in Enterprise System

Given

- Attachment = **10 MB**
- PGP overhead = **800 bytes**
- Emails/day = **5000**

Assume

1 MB = 1,048,576 bytes

10 MB

= $10 \times 1,048,576$

= **10,485,760 bytes**

A. Total Data per Day

Each email

= 10,485,760 + 800

= **10,486,560 bytes**

Daily data

= $10,486,560 \times 5000$

= **52,432,800,000 bytes**

≈ **52.43 GB/day**

B. Percentage Storage Increase

= **0.0076%**

C. Computational Overhead

Encryption requires

- RSA encrypts session key
- AES encrypts attachment
- Digital signature generation
- Signature verification
- Hash computation

The RSA and signature operations consume much more CPU than the AES encryption.

D. Optimization

- Compress attachments before encryption.
- Batch email processing.
- Hardware cryptographic accelerators.
- Multi-threaded encryption.
- ECC instead of RSA where supported.
- Cache public keys.

3. IPSec Transport vs Tunnel Mode Performance Comparison

Given

Original payload = **1000 bytes**

IP header = **20 bytes**

Original packet

= **1020 bytes**

Transport overhead = **30 bytes**

Tunnel overhead = **50 bytes**

Traffic

= **20,000 packets/sec**

A. Packet Size

Transport Mode

$1020 + 30$

= **1050 bytes**

Tunnel Mode

$1020 + 50$

= **1070 bytes**

B. Bandwidth per Second

Transport

$1050 \times 20,000$

= **21,000,000 bytes/sec**

$\approx$ **21 MB/s**

Tunnel

$1070 \times 20,000$

= **21,400,000 bytes/sec**

$\approx$ **21.4 MB/s**

C. Comparison

Feature	Transport Mode	Tunnel Mode
Overhead	Lower	Higher
Security	Payload only	Entire packet
Speed	Faster	Slightly slower
VPN Support	Limited	Excellent
Bandwidth	Lower	Higher

D. Recommendation

Internal Communication

Use **Transport Mode**

Reason

- Lower overhead
- Better performance
- Suitable for trusted networks

External Communication

Use **Tunnel Mode**

Reason

- Encrypts entire packet
- Better privacy
- Standard VPN implementation

4. S/MIME Certificate Management

Given

Employees

= **12,000**

Certificate size

= **2 KB**

Renewal overlap

= **Old + New**

A. Storage During Renewal

Each employee stores

2 KB + 2 KB

= **4 KB**

Total

= $12,000 \times 4 \text{ KB}$

= **48,000 KB**

≈ **46.88 MB**

B. Distribution Overhead

New certificates

$12,000 \times 2 \text{ KB}$

= **24,000 KB**

≈ **23.44 MB**

Therefore, approximately **23.44 MB** must be distributed across the network during renewal.

C. Challenges

- Certificate renewal scheduling
- Certificate revocation
- Expired certificates
- User key management
- Secure storage
- PKI scalability

- Administrative complexity

D. Optimization

- Centralized PKI
- Automated certificate renewal
- Certificate Lifecycle Management software
- Short-lived certificates
- OCSP for revocation checking
- Automated backups

5. Email Encryption Latency in Cloud-Based Systems

Given

AES time

= **0.2 ms/MB**

Attachment

= **8 MB**

RSA key exchange

= **5 ms/email**

Emails/hour

= **8000**

A. Encryption Time per Email

AES

= 8×0.2

= **1.6 ms**

RSA

= **5 ms**

Total

= $1.6 + 5$

= **6.6 ms/email**

B. Processing Time per Hour

6.6×8000

= **52,800 ms**

= **52.8 seconds/hour**

C. Performance Bottleneck

AES

= **1.6 ms**

RSA

= **5 ms**

RSA consumes approximately **76%** of the total encryption time, making the asymmetric key exchange the primary bottleneck.

D. Optimization

- Replace RSA with **ECC** for faster key exchange.

- Use hardware security modules (HSMs).
- Parallelize AES encryption.
- Use AES-NI CPU instructions.
- Batch processing for multiple emails.
- Reuse secure sessions when appropriate.

Final Answers Summary

Question	A	B	C
1. SSL/TLS	23.32 MB/s	14.24% overhead	3.07 MB/s handshake bandwidth
2. PGP	52.43 GB/day	0.0076% storage increase	RSA/signatures are the major computational overhead
3. IPSec	Transport: 1050 B, Tunnel: 1070 B	21 MB/s vs 21.4 MB/s	Transport for internal, Tunnel for external
4. S/MIME	46.88 MB storage	23.44 MB distribution	Automation and centralized PKI recommended
5. Email Encryption	6.6 ms/email	52.8 s/hour	RSA is the main performance bottleneck